Literal presentations and local periods in hexagonal extensions

An application to Fernandes–Leemans–Ivić Weiss

The retained primary is the 21-page author edition of *Hexagonal extensions of toroidal maps and hypermaps* [1]. We preserve its parameter order, literal presentations and regular/chiral conventions. The original constructions and questions are attributed to its authors. All group orders below are full *type-preserving* automorphism orders; correlations are separate.

The complete earlier marked-kernel common paper and the complete new *Rotation certificates and affine period families* supplement are paid premises. The latter contains the entire accepted cubic, square and exact finite proofs formerly located in the L121 application, now shared with this application. No sibling application proof is free. The standard torus and faithful affine Coxeter conventions remain explicit classical inputs. We do not reprove the prior classifications quoted as source Tables 1–2 or Theorems 3.1–3.3.

1 One certificate for each literal finite row

For each parameter row let $P = \langle x, y, z \mid \mathcal{R} \rangle$ be exactly the source's rotation presentation. The entire six presentation builders occur in the required program `presentation_engine.g`. Parameters are read from `table_roster.json`, not reconstructed from a claimed order. Put $a_0 = 1, a_1 = x, a_2 = y, a_3 = z$ and $P_i = \langle a_j^{-1} a_k : j, k \neq i \rangle$. The engine establishes an exact isomorphism from the abstract presentation to its permutation group. This is an upper-bound computation as well as a finite image. In T8.8, complete coset enumeration over $\langle y, z \rangle$ gives index 17280; the abstract local presentation has order 150. The resulting permutation image has order 2592000, equal to $17280 \cdot 150$, so the upper and lower bounds coincide. ACE is bounded by 500000 cosets,5000000 workspace words and 60 seconds. The first general attempt was stopped at 120 seconds; its receipt is not treated as evidence against finiteness.

For every row the engine checks all 256 intersections of the 16 pair-difference subgroups, all 16 normalized flag counts, generation for each of the four rank-three residues and the six actual rank-two residue incidence graphs. The shared theorem gives thinness, residual connectivity and precisely two rotation-group chamber orbits. If simultaneous inversion satisfies every literal relator it is an involutory automorphism. The faithful two-copy semidirect completion is checked for all 256 marked intersections and 16 coset-Helly identities. Otherwise a specified failed inverse relator certifies chirality and that P itself is the full type-preserving group.

Theorem 1 (The computed literal rows)**.** *All 66 finite rows below define finite regular or chiral hypertopes, with the exact full orders and actual marked local groups displayed. Each has the source's asserted regular/chiral status. The full orders in source Table 5 rows 8 and 10 must be doubled for the literal presentations in this retained edition.*

Proof. Apply the shared rotation and local-kernel propositions to the exact per-row outputs. The abstract upper bound and generator maps are separate from the subgroup and flag tests. Each local group is identified by an onto map from its complete two-generator presentation with matching order; its full translation kernel is also computed. The regular completion has order $2|P|$ and the chiral full group has order $|P|$. No assertion is transferred between rows merely because their numerical orders agree. The required software and entire row roster are explicit computational premises. □

R and C mean regular and chiral. The four local orders are rotation orders in omitted-type order 0,1,2,3. Seven inherited-family rows are handled separately. Printed structural descriptions are audited separately below. They are never identified merely from an order.

Row	Literal parameters	Printed g	Proved g	Type	Local rotation orders
T3.1	1, 2; 1, 2	1,344	1,344	C	12/42/12/42
T3.2	1, 2; 2, 1	20,160	20,160	C	12/42/12/42
T3.3	2, 0; 1, 2	672	672	C	12/42/12/24
T3.4	2, 0; 1, 3	4,368	4,368	C	12/78/12/24
T3.5	2, 0; 1, 4	16,128	16,128	C	12/126/12/24

Row	Literal parameters	Printed g	Proved g	Type	Local rotation orders
T3.6	$2,0;\ 2,3$	13,680	13,680	C	12/114/12/24
T4.1	$2,0;\ 2,0$	240	240	R	6/24/6/24
T4.2	$2,0;\ 6,0$	720	720	R	6/72/6/24
T4.3	$3,0;\ 3,0$	2,916	2,916	R	6/54/6/54
T4.4	$3,0;\ 4,0$	241,920	241,920	R	6/96/6/54
T4.5	$3,0;\ 1,1$	324	324	R	6/18/6/54
T4.6	$3,0;\ 2,2$	41,472	41,472	R	6/72/6/54
T4.7	$1,1;\ 1,1$	108	108	R	6/18/6/18
T4.8	$2,2;\ 2,2$	13,271,040	13,271,040	R	6/72/6/72
T4.9	$3,0;\ 1,3$	33,696	33,696	C	6/78/6/54
T4.10	$4,0;\ 1,2$	12,096	12,096	C	6/42/6/96
T4.11	$6,0;\ 1,2$	756,000	756,000	C	6/42/6/216
T4.12	$1,2;\ 3,6$	2,016	2,016	C	6/126/6/42
T4.13	$1,2;\ 4,4$	36,288	36,288	C	6/288/6/42
T4.14	$1,2;\ 6,6$	2,268,000	2,268,000	C	6/648/6/42
T4.15	$3,5;\ 2,1$	672	672	C	6/42/6/42
T4.16	$1,4;\ 2,1$	2,016	2,016	C	6/42/6/126
T4.17	$1,2;\ 1,2$	672	672	C	6/42/6/42
T5.1	$2,0;\ 2,0;\ 2,0$	240	240	R	6/24/24/12
T5.2	$2,0;\ 3,0;\ 3,0$	1,296	1,296	R	6/54/54/12
T5.3	$2,0;\ 4,0;\ 4,0$	15,360	15,360	R	6/96/96/12
T5.4	$2,0;\ 6,0;\ 2,2$	2,880	2,880	R	6/72/72/12
T5.5	$3,0;\ 2,0;\ 2,0$	1,296	1,296	R	6/24/24/27
T5.6	$3,0;\ 2,0;\ 3,0$	13,824	13,824	R	6/24/54/27
T5.7	$3,0;\ 2,0;\ 4,0$	165,888	165,888	R	6/24/96/27
T5.8	$3,0;\ 2,0;\ 5,0$	2,592,000	5,184,000	R	6/24/150/27
T5.9	$3,0;\ 2,0;\ 2,2$	3,888	3,888	R	6/24/72/27
T5.10	$3,0;\ 2,0;\ 4,4$	248,832	497,664	R	6/24/288/27
T5.11	$3,0;\ 3,0;\ 1,1$	972	972	R	6/54/18/27
T5.12	$3,0;\ 1,1;\ 1,1$	324	324	R	6/18/18/27
T5.13	$3,0;\ 1,1;\ 3,3$	2,916	2,916	R	6/18/162/27
T5.14	$3,0;\ 2,2;\ 2,0$	3,888	3,888	R	6/72/24/27
T5.15	$4,0;\ 2,0;\ 2,0$	15,360	15,360	R	6/24/24/48
T5.16	$2,0;\ 1,2;\ 2,1$	336	336	C	6/42/42/12
T5.17	$2,0;\ 1,3;\ 2,5$	2,184	2,184	C	6/78/78/12
T5.18	$2,0;\ 1,4;\ 1,2$	336	336	C	6/42/42/12
T5.19	$2,0;\ 1,4;\ 3,6$	8,064	8,064	C	6/126/126/12
T5.20	$2,0;\ 2,3;\ 3,2$	6,840	6,840	C	6/114/114/12
T5.21	$3,0;\ 1,2;\ 1,2$	275,562	275,562	C	6/42/42/27
T6.1	$2,0;\ 2,0$	384	384	R	4/24/24/12
T6.6	$3,0;\ 1,1$	144	144	R	4/18/18/12
T6.7	$6,0;\ 1,1$	576	576	R	4/72/18/12
T6.8	$3,0;\ 1,3$	58,968	58,968	C	4/54/78/12
T6.9	$1,2;\ 1,2$	2,688	2,688	C	4/42/42/12
T6.10	$1,2;\ 2,1$	1,008	1,008	C	4/42/42/12
T6.11	$1,2;\ 3,1$	58,968	58,968	C	4/42/78/12
T7.1	$3;\ 3,0;\ 1,1$	360	360	R	12/12/18/18
T7.2	$3;\ 6,0;\ 1,1$	23,040	23,040	R	12/12/18/72
T7.3	$3;\ 1,2;\ 1,1$	1,512	1,512	C	12/12/18/42
T7.4	$3;\ 1,4;\ 1,1$	90,720	90,720	C	12/12/18/126
T7.5	$3;\ 2,4;\ 1,1$	774,144	774,144	C	12/12/18/168
T7.6	$4;\ 2,0;\ 1,1$	4,320	4,320	R	24/24/18/24
T7.7	$4;\ 1,1;\ 1,1$	3,456	3,456	R	24/24/18/18
T8.4	$2,0;\ 1,1;\ 4,0;\ 3,0$	165,888	165,888	R	96/54/18/24
T8.5	$3,0;\ 2,0;\ 1,1;\ 2,0$	1,296	1,296	R	18/24/24/54
T8.6	$3,0;\ 2,0;\ 1,1;\ 3,0$	13,824	13,824	R	18/54/24/54
T8.7	$3,0;\ 2,0;\ 1,1;\ 4,0$	165,888	165,888	R	18/96/24/54
T8.8	$3,0;\ 2,0;\ 1,1;\ 5,0$	5,184,000	5,184,000	R	18/150/24/54
T8.9	$2,0;\ 1,2;\ 1,1;\ 6,0$	15,120	15,120	C	18/72/42/24
T8.10	$2,0;\ 1,2;\ 1,2;\ 2,0$	352,800	352,800	C	42/24/42/24
T8.11	$2,0;\ 1,2;\ 1,1;\ 0,3$	7,620,480	7,620,480	C	18/504/42/24

2 Actual local periods and corrections

The standard torus presentations have orders $6(a^2+ab+b^2)$, or $3(a^2+ab+b^2)$ for the triangular hypermap. Twelve ordinary period relations become stronger local quotients in the global presentation. For instance, T7.1 has actual local rotation order 18 where $(3,0)$ would give 54. Each exact local kernel is recorded below, in the ordered translation words of the shared local-kernel proposition. It specifies the marked quotient, not an unlabelled torus selected by order.

Row	Omitted type	Nominal local order	Actual order	Exact kernel basis
T4.2	1	216	72	$(6,0);(2,2)$
T4.12	1	378	126	$(21,0);(4,1)$
T4.15	3	294	42	$(7,0);(2,1)$
T5.4	1	216	72	$(6,0);(2,2)$
T5.17	2	234	78	$(13,0);(3,1)$
T5.18	1	126	42	$(7,0);(2,1)$
T5.19	2	378	126	$(21,0);(4,1)$
T6.6	1	54	18	$(3,0);(1,1)$
T6.7	1	216	72	$(6,0);(2,2)$
T7.1	3	54	18	$(3,0);(1,1)$
T7.2	3	216	72	$(6,0);(2,2)$
T8.9	1	216	72	$(6,0);(2,2)$
T8.11	1	not a period	504	$(42,0);(8,2)$

There is an additional literal issue in source Section 8, physical p.17. Its final h-word is $y^{-1}(yz)^2 = zyz$, whose cube is already trivial from $y^3 = z^2 = 1$. For T8.11, $v = (0,3)$ thus adds no period. The literal global group nevertheless has the printed order 7620480. Its actual local rotation order is 504, with kernel basis $(42,0),(8,2)$ and determinant 84. We certify this actual quotient from the three triangle relations and its full lattice kernel, without inserting a likely intended different word. All other positive Table 8 rows have $h = 0$.

T5.8 has rotation order 2592000 and regular full order 5184000; T5.10 has rotation order 248832 and regular full order 497664. Both pass every intersection, flag and actual-residue test. The printed full totals are 2592000 and 248832. Their structural abbreviations cannot be asserted unchanged when their implied orders conflict; the exact marked presentations and local models specify the corrected objects.

Adding the actual local kernel relations changes no global group, since they already hold in it. Conversely the original period relations hold in each exact local quotient. Thus the same global group is universal for these *actual* local presentations. A claim with uncollapsed preassigned periods is stronger and is not made for the listed exceptions.

3 Shared affine families and inherited finite rows

For Table 6 rows 2–5 and all $s \geq 3$, the complete shared cubic proof gives E/sD of full order $48s^3$, its marked residues and a faithful degree-$4s$ CPR representation. In its notation (a,b,d,c) use

$$x = ba, \qquad y = bc, \qquad z = bd.$$

Then $(y^{-1}z^{-2})^2 = (c(bd)^2b)^2$ is the fixed local relation, and $(x^{-1}z^{-2})^s = (a(bd)^2b)^s$ is the variable relation. The ordered parameter pair is $((2,0),(s,0))$. The full cubic toroid has order $96s^3$; the common proof retains its explicit index-two embedding and the source CPR labels. The endpoint $s = 2$ is separately certified as T6.1, not inferred from the faithful-$4s$ argument.

Table 8 row 1 is the entire shared square family with parameters $((2,0),(1,1),(s,s),(2,0))$ and full order $1152s^4$, for $s \geq 1$. The fixed affine group, integral parabolic lattices, period kernel, intersections and flag transitivity are proved in the paid common proof. Table 8 rows 2–3 are the complete shared finite cases of orders 720 and 46080. Their actual first local period has the degree-three quotient proved there, rather than the uncollapsed $(3t,0)$ type. The shared orientation-adapter proof gives exact rotation/reflection presentation identities and explicit marker permutations alongside the unchanged proof migration; matching orders alone is not the argument.

4 Structural columns and corrected extension claims

Every one of the 73 printed group descriptions is retained in the ledger. V denotes one of the 29 previously verified named isomorphisms. F denotes one of the 35 complete recursive normal-subgroup, quotient and

complement certificates in the whole structural companion [5]. S denotes one of the seven shared affine or finite structural proofs in [4]; O records a printed order incompatible with the literal universal group. The two complete companion proofs and their whole required programs and input data are part of this proof boundary. An unnamed action does not remove the obligation to prove normality, the quotient, or a stated splitting. Conversely, verifying a bracketed extension shape does not claim it uniquely identifies all abstract groups with those factors.

Thirty-four F rows verify the printed shape under explicit brackets. For T4.8, the asserted split stage must instead be dotted:

$$\big((2^{5+6}.(A_5 \times C_3)) : S_3\big) : S_3.$$

The companion supplies both the corrected positive shape and the obstruction to the printed splitting. Its perfect terminal derived subgroup of order 120 cannot have an index-two subgroup isomorphic to A_5; the complete subgroup argument is retained there. The seven S rows have explicit normal factors and complements, including the uniform square-family splitting for every positive parameter. No new geometry is inferred solely from these structural descriptions.

The required `structural_groups.py` constructs the same full group from the certified rotation generators and the checked inversion automorphism. For the 29 V rows it constructs the stated target and verifies an actual bijective group homomorphism. It records the source generators, their images and the target generators. The target denoted $A \times_{C_2} S_3$ is the subgroup of $A \times S_3$ where the two quotient signs agree, using $A/A' = C_2$; this is the "fibre" notation below. The group $\mathrm{P\Sigma L}_2(27)$ is the preimage of the order-three subgroup of $\mathrm{Aut}(\mathrm{PSL}_2(27))/\mathrm{PSL}_2(27)$. For $\mathrm{P\Sigma L}_2(49)$ the program explicitly adjoins the field map $u \mapsto u^7$ to the fractional-linear $\mathrm{PSL}_2(49)$ action. This distinguishes the field action from the inequivalent diagonal extension $\mathrm{PGL}_2(49)$. Wreath products use their full stated permutation action. Classical finite-group constructors and exact isomorphism algorithms are declared software premises; no source extension action is inferred from composition factors.

For T8.11 a smaller factor certificate replaces an expensive direct isomorphism search. The two minimal normal subgroups have orders 2520 and 504. Their opposite centralizers have orders 5040 and 1512, commute, intersect trivially, and generate the full group. Separate bijections identify them with S_7 and $\mathrm{Aut}(\mathrm{PSL}_2(8))$. Multiplication therefore proves the displayed direct-product identification. The first bounded direct search timed out; no failure claim was inferred.

Row	Printed group column	Exact scope here
T3.1	$2 \,.\, L_2(7) : 2 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T3.2	$2 \,.\, A_7 : 2 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T3.3	$L_2(7) : 2 \times 2$	V: PGL(2,7) x C2
T3.4	$L_2(13) \times 2 \times 2$	V: PSL(2,13) x C2 x C2
T3.5	$Q_8 \,.\, (L_2(7) \times 2) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T3.6	$L_2(19) : 2 \times 2$	V: PGL(2,19) x C2
T4.1	$A_5 : 2 \times 2$	V: S5 x C2
T4.2	$S_5 \times S_3$	V: S5 x S3
T4.3	$3^{1+2} \times 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.4	$\mathrm{Aut}(L_3(4))$	V: Aut(PSL(3,4))
T4.5	$3^{1+2} : 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.6	$2^{1+6} : 3^2 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.7	$3^{1+2} : 2^2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.8	$2^{5+6} : A_5 \times 3 : S_3 : S_3$	F: corrected nonsplit stage; complete companion proof.
T4.9	$L_3(3) : S_3$	V: Aut(PSL(3,3)) fibre_C2 S3
T4.10	$U3(3) : 2$	V: Aut(PSU(3,3))
T4.11	$U3(5) : S_3$	V: Aut(PSU(3,5))
T4.12	$2 \,.\, L_2(7) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.13	$U3(3) : S_3$	V: Aut(PSU(3,3)) fibre_C2 S3
T4.14	$3 \,.\, U3(5) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.

Row	Printed group column	Exact scope here
T4.15	$2 \,.\, L_2(7) : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.16	$2 \,.\, L_2(7) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.17	$2 \,.\, L_2(7) : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.1	$S_5 \times 2$	V: S5 x C2
T5.2	$S_3 \times S_3 \times S_3 : S_3$	V: S3 wreath S3
T5.3	$2^{1+6} : (A_5 : 2)$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.4	$S_5 \times 2^2 : S_3$	V: S5 x S4
T5.5	$6 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.6	$2^{1+6} : 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.7	$2^{1+6} : 6 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.8	$2 \,.\, (A_5 \times (A_5 \times A_5)) : 3 : 2$	O: printed order incompatible; no structural identification asserted.
T5.9	$S_3 \times 3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.10	$2^{1+6} : (3 : S_3 : S_3 : 3) : S_3$	O: printed order incompatible; no structural identification asserted.
T5.11	$3^{1+2} : 6 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.12	$3^2 \times S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.13	$3^{1+2} \times 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.14	$S_3 \times 3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.15	$4 \,.\, (2^4 : A_5) : 2 \times 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.16	$L_2(7) : 2$	V: PGL(2,7)
T5.17	$L_2(13) \times 2$	V: PSL(2,13) x C2
T5.18	$L_2(7) : 2$	V: PGL(2,7)
T5.19	$Q_8 \,.\, L_2(7) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.20	$L_2(19) : 2$	V: PGL(2,19)
T5.21	$3^{2+6} : 7 : 3 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.1	$2^{1+4} \times 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.2	$S_3 \times S_3 \times S_3 : S_3$	S: complete shared normal-factor and complement proof.
T6.3	$2^{1+6} : 2^2 : S_3$	S: complete shared normal-factor and complement proof.
T6.4	$5^3 : 2 : 2 : 2 : S_3$	S: complete shared normal-factor and complement proof.
T6.5	$2^3 : S_3 : S_3 : S_3 : S_3$	S: complete shared normal-factor and complement proof.
T6.6	$S_3 \times 2^2 : S_3$	V: S3 x S4
T6.7	$2^4 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.8	$L_2(27) : 3 : 2$	V: Aut(PSL(2,27))
T6.9	$2^6 : 7 : 3 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.10	$L_2(7) \times 3 : 2$	V: PGL(2,7) x C3
T6.11	$L_2(27) : 3 \times 2$	V: PSigmaL(2,27) x C2
T7.1	$A_5 : S_3$	V: S5 fibre_C2 S3
T7.2	$2^4 : A_5 : 2 : 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T7.3	$L_2(8) : 3$	V: Aut(PSL(2,8))

Row	Printed group column	Exact scope here
T7.4	$L_2(8) \times A_5 : 3$	V: Aut(PSL(2,8)) x A5
T7.5	$2^9 \,.\, L_2(8) : 3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T7.6	$A_6 : 2 \times S_3$	V: S6 x S3
T7.7	$2^{1+4} : 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.1	$s^4 : 2^{1+4} : S_3 : S_3$	S: complete shared normal-factor and complement proof.
T8.2	$A_5 \times 2 : S_3$	S: complete shared normal-factor and complement proof.
T8.3	$2^4 : A_5 : 2 \times 2 : 2 : S_3$	S: complete shared normal-factor and complement proof.
T8.4	$2^{1+6} : S_3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.5	$S_3 \times S_3 \times S_3 : S_3$	V: S3 wreath S3
T8.6	$2^{1+6} : 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.7	$2^{1+6} : S_3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.8	$2 \,.\, (A_5 \times (A_5 \times A_5)) : 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.9	$3 \times S_7$	V: C3 x S7
T8.10	$L_2(49) \times 3 : 2$	V: PSigmaL(2,49) x C3
T8.11	$S_7 \times (L_2(8) : 3)$	V: S7 x Aut(PSL(2,8))

5 The diagram list and its limit

The ten patterns in source Figure 1 represent sixteen fixed edge-labelled cases after expanding its parameter $p \in \{3, 4, 5, 6\}$. Here is the complete elementary reduction. A connected rank-four nonlinear diagram has one of five underlying graphs: a star, a triangle with a pendant edge, a four-cycle, K_4 minus one edge, or K_4. Every rank-three triangle must be spherical or Euclidean, and at least one must be Euclidean. The retained triangle criterion is $1/p + 1/q + 1/r \geq 1$, with equality for a toroidal face and omitted edges assigned label two. An edge with label greater than six would have a neighbouring edge of label at least three by connectedness, violating this criterion. Thus labels two through six suffice.

A star must have arm labels $(3, 3, 6)$, $(3, 4, 4)$ or $(4, 4, 4)$. A triangle with a pendant edge has all triangle labels three and pendant label three through six. In a four-cycle, a label six forces its neighbours to be three and permits the opposite label three through six. With no six, an affine adjacent pair must be $(4, 4)$; labels five cannot then occur. The possibilities are an adjacent pair of fours and two threes, three fours and one three, or four fours. Finally, in K_4 minus an edge and in K_4, all edges belong to triangles and hence all have label three. This gives respectively $3 + 4 + 7 + 1 + 1 = 16$ cases, exactly the source's ten parametrized patterns. The optional rational enumeration in `diagram_types.py` checks this case split; it is not a premise of the written argument. The reduction is a list of possible diagrams, not a proof that every case has a finite universal hypertope.

6 The isolated nonorientable example

The complete separate note [6] verifies the source's pp. 18–19 assertion about $\{\{4, 4\}_{(4,0)}, \{4, 6\}_3\}$. The literal universal group has order 768 and local orders 128 and 48. All 256 marked subgroup intersections and the local universal maps are checked. The vertex figure has face counts $(4, 12, 6)$ and a closed flag walk of odd length nine, so it has nonorientable genus four. The entire two-page argument and its exact finite-presentation program are required parts of this application; the displayed permutation image alone is not used as a universal upper bound. This is verification of an isolated source assertion, not an answer to a subsequent open question.

7 Full source boundaries and questions

Section 2's coset construction and C-plus terminology retain Tits and the earlier-literature attributions. The shared theorem supplies the finite rotation-to-geometry direction with its actual hypotheses. Source Tables

1–2 and Theorems 3.1–3.3 quote earlier classifications and remain retained inputs. Diagram (4)'s asserted presentation reduction is distinguished from diagram (5)'s explicitly experimental existence claim. The complete correlation companion [7] proves inverse presentation maps and both parameter transformations for every positive integer parameter. For each existing regular universal fork hypertope, its actual prescribed residues give the corresponding universal string polytope and recover the fork marking. Its correct rank-four generator list contains the fourth generator omitted in the source parenthesis. For diagram (5), the exact group and conditional intersection-property claims are proved, while the experimental flag-transitivity and existence claim remains qualified. There is no unrestricted converse from a parity character alone. The whole four-page companion is part of this application. Bounded failed searches in Section 8 are documentary, not nonexistence theorems. The source's principal finite tables concern the first three diagrams.

Problems 9.1–9.2 and Conjectures 9.1–9.3 are separate from positive-row recovery. Previously reviewed Problem 9.2 work constructs finite quotient geometries for every integer parameter and proves the literal universal group infinite for all parameters at least six. Its exact old proofs and reviews are retained separately; no fresh solution, priority or current-open status is claimed. The four small universal orders now have the T8.5–T8.8 certificates. Finite quotients do not imply finite universal groups. None of these computations proves the source lists exhaustive or proves its negative conjectures.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, retained 21-page author edition (2018 institutional file).

[2] *Rotation certificates and affine period families*, complete supplement to *Supported kernels, translation lattices and finite marked quotients*, private proof migration.

[3] M. E. Fernandes, D. Leemans, C. A. Piedade and A. Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, twelve-page author edition.

[4] *The seven shared group structures in the hexagonal tables*, complete accepted shared structural companion, four pages.

[5] *Checking the structural descriptions of finite hexagonal extensions*, complete accepted structural companion, four pages.

[6] *A locally toroidal polytope with a nonorientable vertex figure*, complete accepted companion, two pages.

[7] *Correlation reductions and the exact square-torus parameters*, complete accepted companion, four pages.